SYZYGY

Mixed realities? Posthuman immigration, staying foreign, and walking worlds into being



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RCS 2026 photographs: Brian Hartley

University of Glasgow | Royal Conservatoire of Scotland | IMMERSIVE ARTS | CELFYDDYDAU YMDROCHOL

Mushrooms, machines, objects

MycoMusic · mushrooms

2022–24

“The machine acted as a mediator, translating activity into a structure legible to humans, while the mushrooms remained entirely foreign: scientifically explicable as communicative (Adamatzky, 2022), yet experientially opaque.” (Sappho, in press)

Mycomusic: https://www.mariasappho.com/collaborations/with-mushrooms/

Chimère · AI

2022–26

“a community-built AI … a network working across Lesotho, South Africa, Switzerland, and the UK to explore how artificial intelligence might act as a mediator between human and more-than-human agencies” chimere.ai

Driving Ms. Chimère

Sappho et al., 2025

Colin Frank · agential objects

agential objects · machine songs

“how objects can be agential in guiding a fluid practice” (Frank, 2022)

“Machines pervade urban environments, yet their sonic performances are little considered.” (Frank, 2026, Sherbrooke)

Sherbrooke Machine Songs

Frank, 2026

We are humans, and we know as humans. (K. Hayles, 2025)

We are never in fact ‘just human’. (Braidotti et al., 2021, p. 41)

We are already machine. (Agüera y Arcas, 2024, p. 17)

What is a world?

a gathering of relations

worlds as enacted, relational and multiple (Escobar, 2017)

Physical
room, floor, Chladni Flower, sand

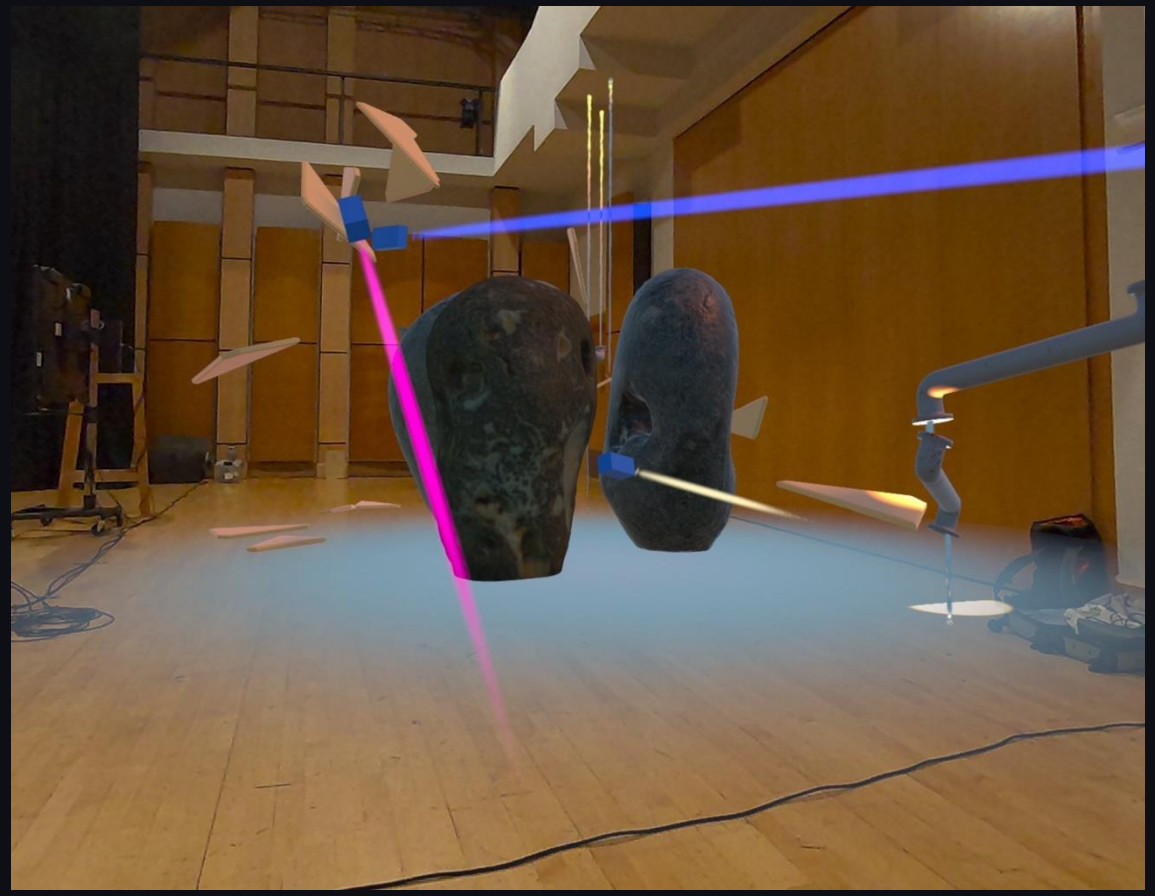
Virtual
pipes, sand, light, hag stones

Sonic
spatialised sound, vibration

Embodied
walking, reaching, proprioception

Social
watching, hesitating, imitating

Machinic
headsets, sensors, Unity, Max, network



Version 2 headset view (RCS, 2026): virtual elements layered onto the physical hall

Version 1
building a world

Version 2
architect & composer

Today
a relay of worlds

Brazil
performance & documentation

VERSION 1 · 2025

Version 1: building a world, physical and digital

VR player · headset

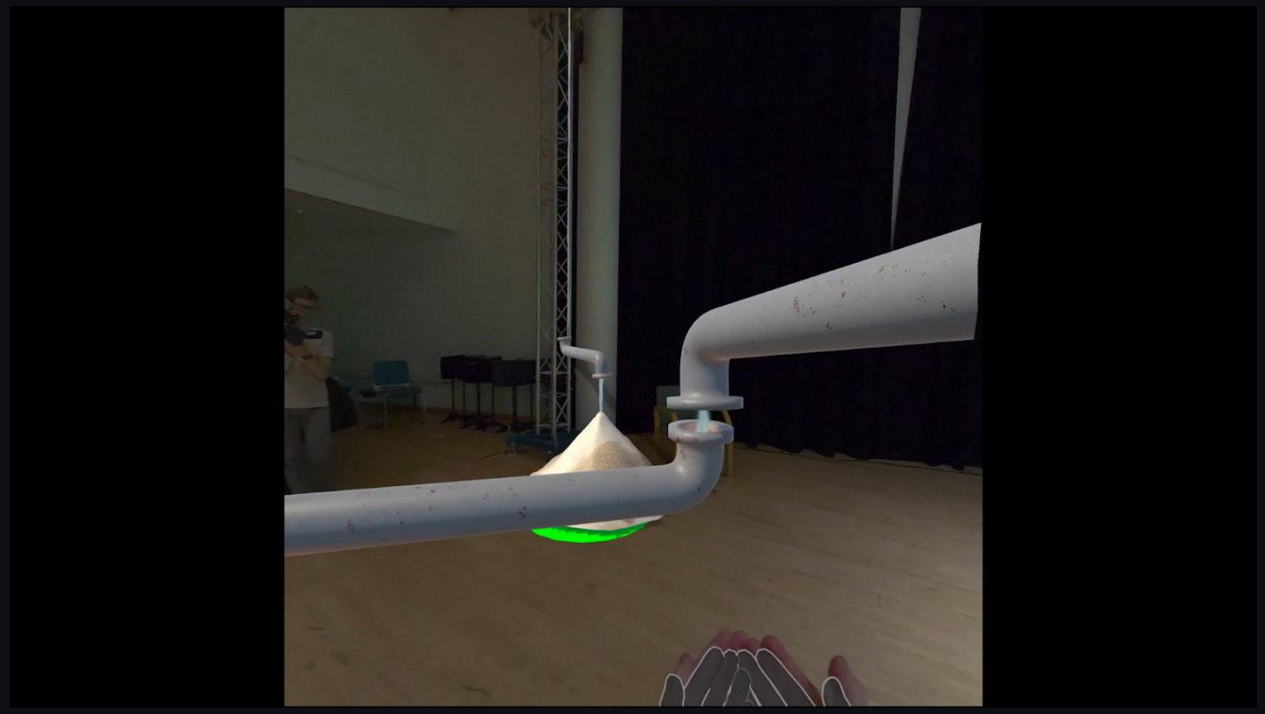
Moves virtual sand through pipes. Sees beyond the physical room, cannot act on physical matter until they find the ‘soluiton’.

Haptic player · handheld device

Senses magnetic fields, sends pitches to the Chladni Flower. Acts on physical matter, never perceives the whole.

Participants as “both microorganism and world-builder” (syzygy, 2025)

A mixed-reality ecosystem for speculative terraforming (Sappho, in press)



Version 1: the Chladni Flower, headset player and handheld player

https://youtu.be/qTpAZd6wQiQ

Audience encounters: discovery, trust, convergence

“I feel like trust was built. Right? Because it’s like, I’m relying on you now.”

Interview 08

“I like this idea that I can affect some things and then you can affect some things and we kind of need… to work out where we are, what we’re doing together. It’s fun.”

Interview 02

“I felt like an infant… playing with blocks and the blocks are fascinating, and they’re brand new, even though they’re really simple.”

Interview 06

Partial knowing invited curiosity, play and trust.

“It’s like really, really beautiful to have an experience where in whatever way, you’re co-operating… by somehow meeting in the middle… these moments of convergence.” (Interview 08)

“Unpredictable encounters transform us; we are not in control, even of ourselves.” (Tsing, 2015, p. 20)

A lovely new book by a good friend on themes of trust: Burke, R., & Onsman, A. (2026). Trust in the Art of Free Improvisation (1st ed.). Routledge. https://doi.org/10.4324/9781003614616

Version 2: co-location and greater agency for change



— One shared mixed-reality world; 2 people in headsets in the same 'room'

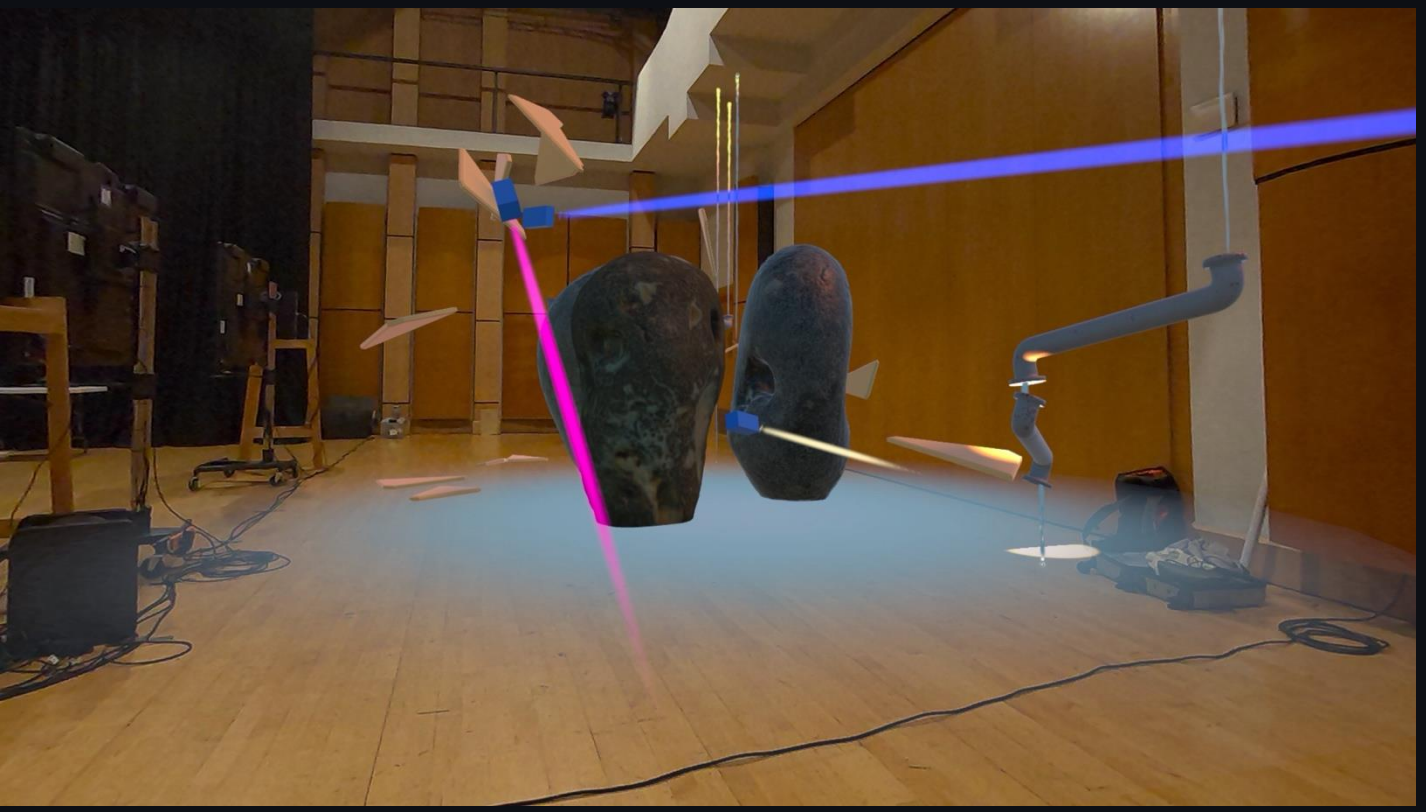
— Pipes, beam emitters and reflector shards can be picked up, moved and re-angledfor direct development of sound-worlds/sculpture

— More human power to reshape the environment



Version 2, Intelligences from the Underground symposium, Royal Conservatoire of Scotland. Photo: Brian Hartley

Absorption, cooperation and the changing role of the world



Version 2, RCS 2026.

WRITTEN FEEDBACK · 4 RESPONSES

“i loved it !!! wish i had more time to really get into it and “compose”” Response 1

“The moment one discovers the installation is too playful and overwhelming (in the best possible way) to really allow team work right away” Response 2

“it was fun, they were building something and that inspired me to imitate (and helped me understand very intuitively what was going on)” Response 3

“I left the installation not because I was told to do so. It was because I wanted to let other people experience it.” Response 2

Coexistence not as fusion but *“the spacing of being-together”* (Ricco, 2015, p. 58)





VERSION 2 · RCS, JUNE 2026

Walking, listening, exploring, designing

“you could walk around it … you were properly in the space rather than sat in a chair being shown something”

“you know where a sound is because your arm is there”

Response 4

Bodies as *“zoe/geo/techno-bodies”* (Braidotti, 2022)

VERSION 2 · PERFORMANCE, JUNE 2026

Version 2 in performance: 3D and physical graphic scores and instruments

— Audience-made 3D compositions from the RCS workshop became graphic scores

— Members of the Glasgow Improvisers Orchestra played them on live instruments, each reading one ray of light and the global changes to the light

— The physical element: my light instruments for interactive sound control, and neon lights made by Raymond MacDonald as graphic scores

— An exploration between 3D and physical graphic scores and gesture based performance.

Spatial 3D score · **Gestural** light instrument · **2D** neon graphic scores



Version 2 in performance, the Old Hairdressers, Glasgow, June 2026

In concert: three modes, one practice



Intelligences from the Underground, the Old Hairdressers, Glasgow, 29 June 2026 · excerpt, 1'30"

Version 2.1: a relay of worlds

1
A pair makes a world: an architecture or score

2
Explain directions for its experience

3
The next pair walks it, and changes it, drastically or not

4
They pass it on with their own directions

By the end of the workshop

a world walked into a certain state, and unfinished at the same time

“tried to get them to stay but they would fade … like bubbles popping”

Version 2, Response 1, on the light beams

“A leaf, a gourd, a sack, a sling, a net, a bag. A recipient.” (Le Guin, 2020, p. 2)



Development views toward London: headset passthrough, domestic test space

Threads we will walk through together

I — Whose world are we building, and what world was already here?

II — How do we act in a world we do not understand?

III — When we can change, what do we owe the world we change?

IV — What do you change, and what do you leave for the next person?

V — When a world keeps changing hands: performance, score, instrument, document?

While pairs walk the world, we follow these threads through the network

THE WORKSHOP · THE RELAY

walk your world into being

1 Design a world

— With your partner, design a structure, a world, an architecture, a score: whatever you make of it

— Leave directions for the next pair: first, how to experience it

2 Walk it, change it, pass it on

— Enter the world the last pair left

— First, experience it as their directions suggest

— Then change it: drastically, or not

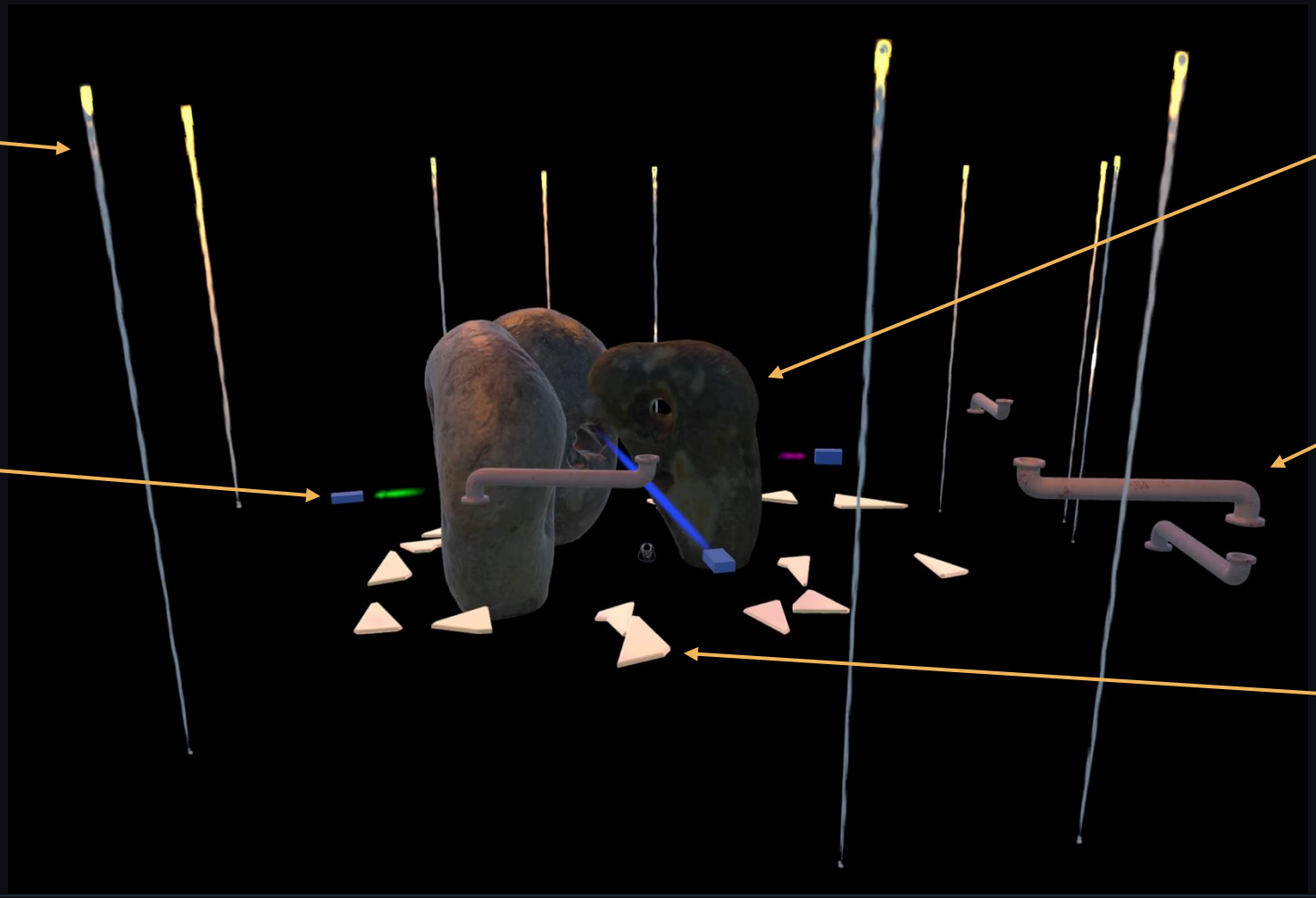
— Leave your own directions for the next pair

We pass worlds on, walking into each other's worlds and changing them. By the end: a world walked into a certain state, and still unfinished.

What’s in the world

Falling sand
(particle system, redistributable)

Neon beam emitters
(moveable, sound objects)



Hag Stones
(Immobile. Produce sound when struck by sand)

Pipes
(Moveable. Redistribute falling sand and sound)

Reflector Shards
(Moveable. For reflecting neon beams to make sound)

THE WORKSHOP · HOW

How to act in the world

Controllers

Recommended. Grab and move objects with the grip and trigger buttons.

Hand tracking

Or put the controllers down: pinch and grab with your hands. Less precise, but possible.

Walk

Move through the room with your body. There is no joystick travel.

Listen as you move

The sound is spatial. Move actively to hear everything: the mix changes as you walk.

Join the relay in pairs

PROVOCATION I · WORLD-BUILDING

Whose world are we building, and what world was already here?

«un mundo donde quepan muchos mundos» · a world where many worlds fit (EZLN, 1996)

ARRIVAL

“This arrival is not into land but into assemblages, machinic, ecological, and affective systems that we do not stand outside of but enter with our bodies, histories, and everyday lives.” (Sappho, in press; Hayles, 2008)

ALREADY LIVED

“posthumanist language does not invent relationality so much as describe practices of belonging and survival that are already lived, particularly within Indigenous and diasporic contexts.” (Sappho, in press; Todd, 2015; Yusoff, 2018)

ALREADY HAPPENED

“These narratives remind us that apocalypses have already taken place, and that the pluriversal futures we imagine are likewise shaped through rupture and survival, emerging in the everyday work of building, grieving, and reimagining our shared worlds.” (Sappho, in press; Dillon, 2012)



THE TROUBLE WITH WORLD-BUILDING

The trouble with world-building

World-making

“If world-making has often carried colonial weight, the imposition of singular authorship onto land, bodies, or futures, then what is needed is closer to world-waking” (Sappho, in press)

Discovery

“The sublime wonder typical in pulp of contact narratives is inverted from an Indigenous perspective, which sees nothing of new worlds discovered, only rupture as the old world is catastrophically changed” (Dillon, 2012, p. 85)

Extraction

The Anthropocene is not a neutral horizon but a terrain marked by racialised extraction and dispossession (Yusoff, 2018)

The standard world

a world that “carries on by itself. People don’t perform it. It’s outside us and we are contained by it” (Escobar, 2017, p. 86)

PROVOCATION II · STAYING FOREIGN

Can we act well in a world we do not understand?

“To step into these assemblages is a kind of migration, inward and relational, that dialogues with, but also departs from, frameworks of geographic displacement, diasporic experience, and posthuman citizenship” (Sappho, in press; Dedeoğlu, 2023; Häkli, 2018)

“The border is a space of collective encounters” (Ozguc & Burridge, 2023)

“belonging is not guaranteed by origin but practiced through relation” (Sappho, in press, after Kimmerer, 2013)

OTHERWISE

Otherwise: concepts for staying foreign

Posthuman immigrant

a continual arrival into emergent modes of relation, responsibility, and coexistence

Staying foreign

foreignness not as lack but as method

Partial knowing, opacity, asymmetry

accepted as methodological conditions

Improvisation

an epistemic practice: a method of arrival

Virtuosity-as-care

responsiveness, reciprocity, and ethical attention within unstable conditions

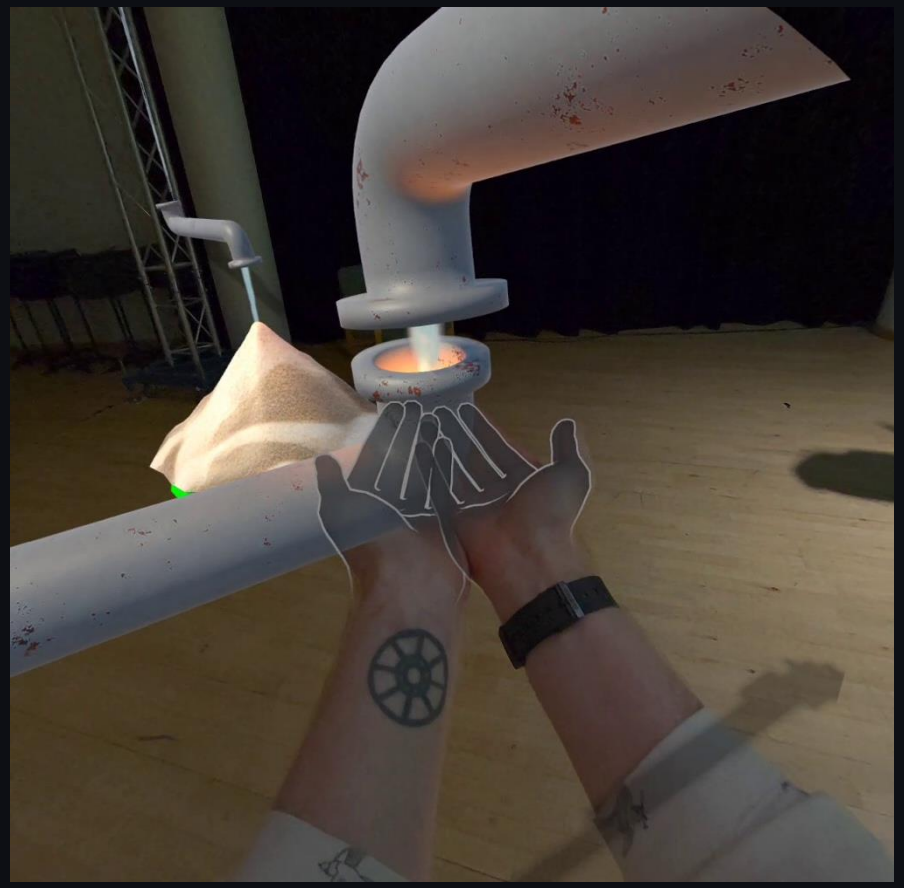
World-waking

attending to fragile, entangled worlds that already exist but are not yet recognised

(Sappho, in press)

VERSION 1 · 2025

Staying foreign: unknowable yet operable worlds



Version 1 headset view: virtual pipes over the physical room

> *“I liked not knowing.”*
>
> Interview 03

> *“Something I really liked and was drawn to was a feeling of revelation.”*
>
> Interview 06

What must be understandable for participation to be possible, and what can remain unknowable?

Foreignness “is not resolved but inhabited as the ground of relation” (Sappho, in press)

“The categories in which we are immersed as forms of life become explicit when you do not quite inhabit them” (Ahmed, 2014, p. 175)

PROVOCATION III · ARCHITECT, TERRAFORMER, COMPOSER

When we can change everything, what do we owe the world we change?

“This means not only designing technology to sit in an otherwise-imperfect world but situating ourselves as otherwise-imperfect designers.” (Keyes, in Drage & McInerney, 2024, p. 99)

“The wish to transcend uncertainty through control and technological mastery is, after all, ‘a colonial fantasy of mastery through technology’” (Sappho, in press; Atanasoski & Amaya Schaeffer, 2024, p. 137)

“Existence is no longer an individual affair” (Barad, 2007, p. xi)

Virtualities

Sappho, 2025

ACROSS VERSIONS

Architect, terraformer, composer

Architect

Conditions for habitation? Condition for exclusion and spatial boundaries?

Thinking here also nods towards omnipotence/religious hierarchy

Terraformer

Eenvironment? Conditions for habitation?

Composer

Sound, space, body, author? Decomoposition a la Haraway?

Overlapping positions

Against “a colonial fantasy of mastery through technology” (Atanasoski & Amaya Schaeffer, 2024, p. 137)

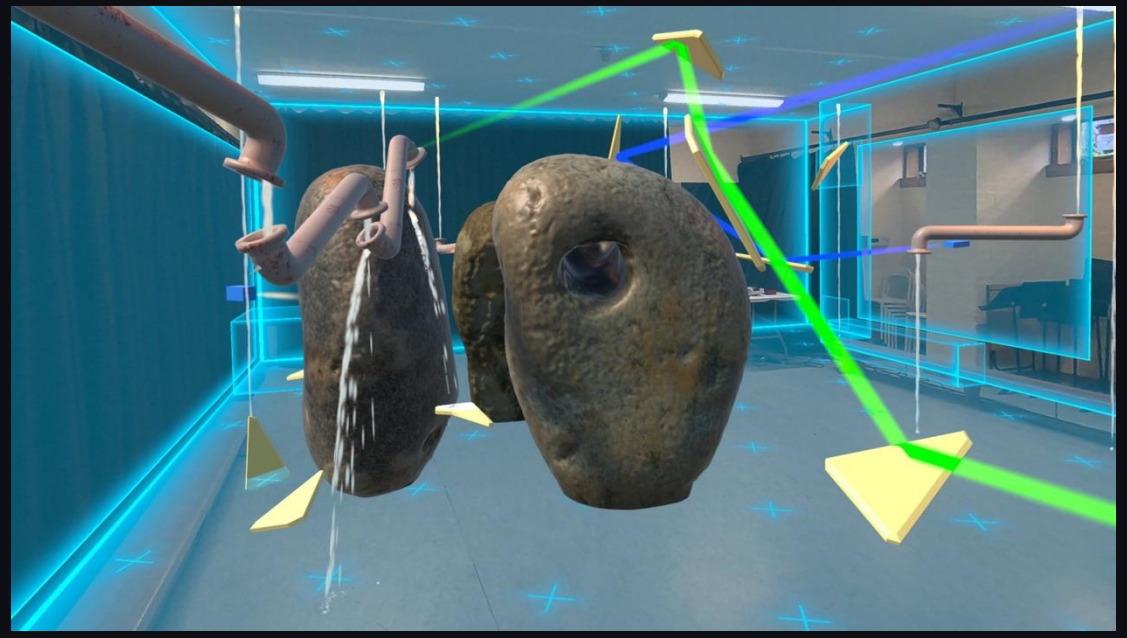
ACROSS VERSIONS

Score? Sculpture?

a spatial arrangement that conditions movement, perception and inhabitation

where and how a body can be

an arrangement of possibilities that another person (maybe) realises in time



Version 2 headset view, RCS 2026

“a composition can be understood as an ecology constructed for a future participant to enter, inhabit, perceive, and alter” (Sappho, in press)

PROVOCATION IV · THE RELAY

What do you change, and what do you leave for the next person?

“As these worlds are shifting, changing, if we are to explore them and notice them, we must accept we are already in great motion, a conceptual nomadism of creative expression and social being” (Sappho, in press, after Lewis, 2021)

“A leaf, a gourd, a sack, a sling, a net, a bag. A recipient.” (Le Guin, 2020, p. 2)

New Directions

MacDonald et al., 2025

now

World-waking: walking worlds into being



Development view toward London, domestic test space

What persists

“the recognition, tending, and invitation into worlds already in motion”
(Sappho, in press)

“Walking the world into being”: a Zapatista formulation (Todd, 2015; Unrau, 2024)

PROVOCATION V · PERFORMANCE AND DOCUMENTATION

When a world keeps changing hands, what is it? a performance, a score, an instrument, a document?

“endowed with capacities to listen, to memorize, to communicate visual information and to synthesize sounds” (Schaub & Bonduki, 2025, on Projetopê)

“a composition can be understood as an ecology constructed for a future participant to enter, inhabit, perceive, and alter” (Sappho, in press)

Projetopê

Schaub & Bonduki, 2025

BRAZIL · PROPOSED

Brazil: the relay as performance and documentation

SCHAUB & BONDUKI, PROJETOPÊ (2025)

A machine "endowed with capacities to listen, to memorize, to communicate visual information and to synthesize sounds"

Each performer's playing "both recorded live and simultaneously displayed in the form of a score"

A guided improvisation between free improvisation and open written composition

(Schaub & Bonduki, 2025)

Is a relayed world a performance, a score, an instrument, a performance object, an experience, a document...?prob all at once.

What should/could a world remember of the people who changed it?

Could listening and memory make the relay legible as it happens?

How do worlds and their directions travel between contexts, how do they travel between worlds?

FUTURE DIRECTION

X-reality: toward reciprocal physical–virtual worlds

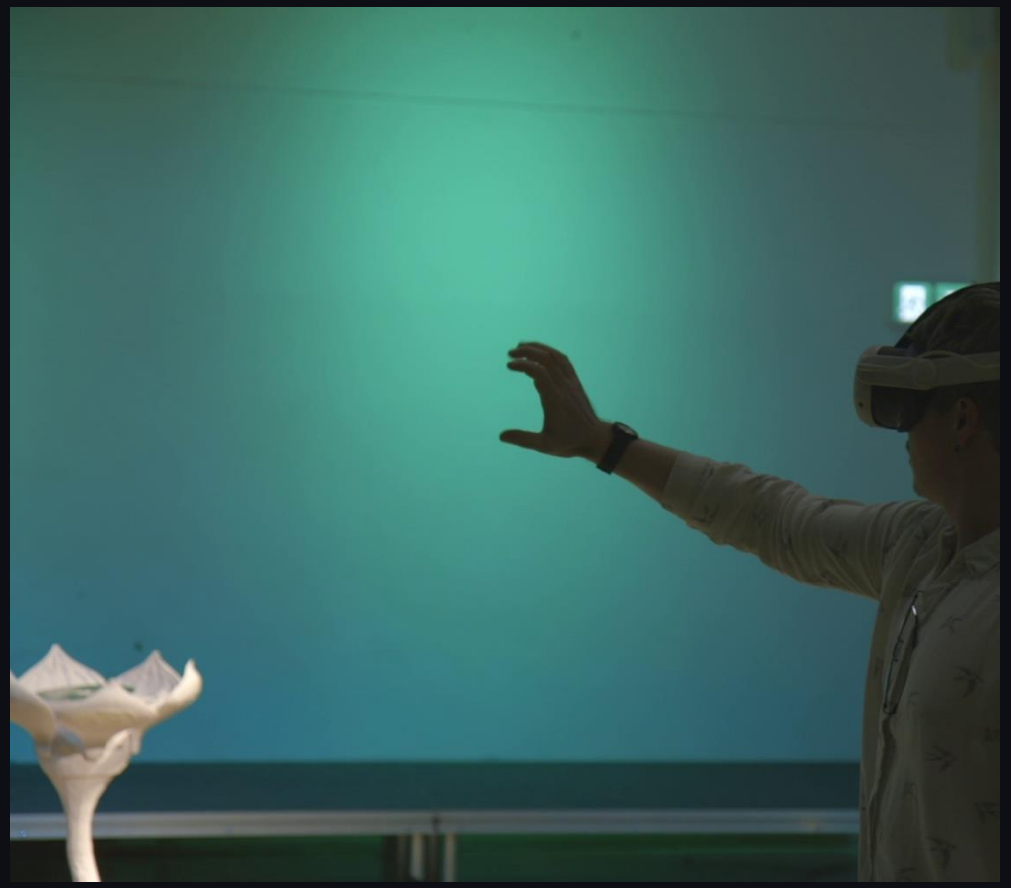
“With the concept of X-reality, I see an end of the virtual or more precisely an end of the binary logic of virtual and real.” (Coleman, 2011, p. 20)

“Rather than having online and off-line ‘selves’, we are now hybrid social beings that can perform elements of our identity across both digital and non-digital spaces.” (Sappho, 2025, pp. 337–338)

“we are empowered via the liberatory task of seizing the digital imaginary as an opportunity, a site to build on and material to build with.” (Russell, 2020, p. 27)

Next plans… bring physical components back into a better-networked relationship with the system, with reciprocal effects between physical processes, virtual environments and actors.

Virtualities
Sappho, 2025



Version 1: the Chladni Flower, a physical component to reconnect

Cognitive assemblages

“We are humans, and we know as humans.” (Hayles, 2025)

“This arrival is not into land but into assemblages, machinic, ecological, and affective systems that we do not stand outside of but enter with our bodies, histories, and everyday lives” (Sappho, in press; Hayles, 2008)

“posthumanism offers language for tracing how these relations now unfold within contemporary technological and algorithmic systems, what Hayles calls cognitive assemblages, in which ‘information, interpretations, and meanings circulate’ through bodies, environments, and networks” (Sappho, in press; Hayles, 2025, p. 99)

where in this world are the assemblages: you, the headset, the stones, the desks, the sound, the person beside you, your memory?

WHILE WE WALK · THREAD

Machine songs, networks, bodies

“The network is the site of the production of knowledge, and the body animates that network. That makes living with creative machines an epistemological practice/project.” (Lewis, 2007, in Sappho, 2025, p. 347)

“Machines pervade urban environments, yet their sonic performances are little considered.” (Frank, 2026)

“We live in an age quickly emerging into a vibrant and complex multi-verse – with social realties, creative practices and tools that are x-panded and powerful” (Sappho, 2025, p. 347)

Sherbrooke Machine Songs
Colin Frank

New Directions
MacDonald et al., 2025

FURTHER READING

SYZYGY

Project page

Posthuman Immigrations

AAWM Journal of Music and Nature (in press)

Driving Ms. Chimère

Community-led work with AI (2025)

Virtualities

Virtues of a techno-socio-creative world (2025)

New Directions

in Musical Collaborative Creativity, OUP (2025)

or find everything at mariasappho.com/research

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